

8.3 Storytelling with charts [↕]

- Watch: Explaining visually to influence with data (part 3) [30:53 mins]

Watch the following video, which expands on the use of storytelling to improve audience engagement and understanding. Different approaches to storytelling, including Aristotle's rhetorical triangle which has been used for thousands of years, are described.

This is useful as it provides deeper understanding and practical advice on how to build a data story and more relatable visualisations.

Build a Slope Chart

A slope chart for the student enrollment data at Opening Horizons

A Chart Title

B Chart Title

C Comparing Student Enrollments for Opening Horizons Locations

Location	Previous Year	Current Year
Salida	395	432
Durango	341	360
Pagosa Springs	302	298
Montrose	198	202
Gunnison	164	150

To create a slope chart in Excel, (A) build a simple line chart, (B) switch rows and columns, and (C) format the chart to its final version.

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James, Graham, Fry & O'Hare, Data Visualisation: Exploring and Explaining with Data, 1st Edition, © 2013 Cengage, All Rights Reserved. Where not be licensed, copied or duplicated, in whole or in part without permission in writing from Cengage.

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Transcript

(<https://learn.usc.edu.au/courses/35242/files/3138654?wrap=1>).

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Explaining visually to influence with data \(part 3\)](https://learn.usc.edu.au/courses/35242/files/3138671?wrap=1) (<https://learn.usc.edu.au/courses/35242/files/3138671?wrap=1>).

Choosing the correct chart to tell your story

To enable effective storytelling with data, it is important to use the correct chart type for both the data and the insights that you are trying to convey. It is useful to try various types of charts before making the final selection.

For example, a bar chart that compares student enrolments at various locations may not tell the story of the data as well as a slope chart. A slope chart is a type of line chart that consists of pairs of connected data points for multiple entities. A slope chart shows the change in each variable over time. Therefore, in the student enrolment scenario, it can visually indicate the increase (positive slope) or decrease (negative slope) in student enrolments over time. It is easy to comprehend and visually supports the storyline.

Using pre-attentive attributes to tell your story

Appropriate use of pre-attentive attributes helps to focus the audience's attention and explain certain insights. This helps to craft and support the story of the data.

For example, colour and colour contrast can be used to emphasise specific data insights, such as the effect of inflation on house prices in a specific area when compared to all other areas.

Size is another pre-attentive attribute that can be used to draw an audience's attention to and explain a specific insight. Using a big associated number (BAN) may effectively convey meaning to an audience. It will also draw the audience's attention and support the story that is being told.

A BAN is simply a number that is displayed in very large font.

□ Watch: Telling stories with data in 3 steps [4.46 mins]

This video provides an introduction to the three elements of a story: set-up, conflict and resolution. The video provides an example of how using these three elements can make a visualisation clearer, sharper and more effective.

As you watch, consider:

- How might this method of thinking about your data influence the way you visualise the data?
- Why does this method create such powerful visualisations?



Source: © Harvard Business Review (2019).

□ Activity: Five megatrends fuelling the rise of data storytelling

Reflect on this webpage about the rise of data storytelling. Note the webpage was published in July 2022.

As you look at the visualisation, consider and make notes on:

- what is good about the visualisation and what is not as good
- whether too much information is being displayed in the visualisation
- what type of storytelling process the visualisation follows
- the cognitive load for the audience.

Explain your reasoning and bring your notes to the workshop.

Routley, N., & Desjardins, J. (2022, July 5). [5 megatrends fueling the rise of data storytelling](https://www.visualcapitalist.com/data-storytelling-megatrends-infographic/)  [\(https://www.visualcapitalist.com/data-storytelling-megatrends-infographic/\)](https://www.visualcapitalist.com/data-storytelling-megatrends-infographic/). *Visual Capitalist*.